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Potassium-40 (^{40}K) is a radioactive isotope that decays through the two modes outlined in Figure 1. In approximately 90% of cases, it undergoes β^- decay to become calcium-40 (^{40}Ca). In the remaining 10% of cases, ^{40}K undergoes electron capture to become argon-40 (^{40}Ar). Collectively, these two processes have a half-life of approximately 1.25 billion years. After 1.25 billion years, 50% of the original amount of ^{40}K remains unchanged, 45% has become ^{40}Ca , and 5% has become ^{40}Ar .

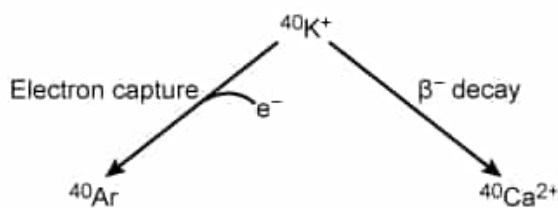


Figure 1 Schematic of the two modes of ^{40}K decay with potassium in the form of a K^+ ion

Based on the known decay rate of ^{40}K , the time that has elapsed since an igneous rock solidified can be determined by measuring the ratio of ^{40}K to its daughter nuclides. However, ^{40}Ca is the most abundant calcium isotope, and igneous rocks commonly contain ^{40}Ca from sources other than the decay of ^{40}K . Consequently, geologists typically only consider the $^{40}\text{K} : ^{40}\text{Ar}$ ratio when determining the age of an igneous rock.

Which of the following is NOT a consideration that must be made when using the $^{40}\text{K} : ^{40}\text{Ar}$ ratio to date an igneous rock?

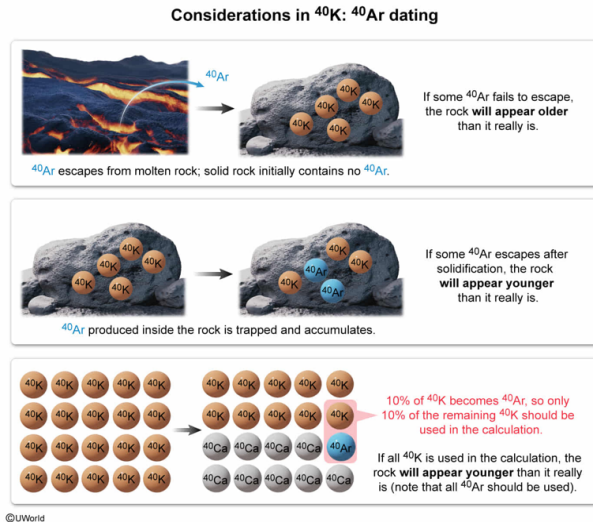
- ☐ A. Only 10% of the ^{40}K remaining in the sample should be used in the calculation. [24%]
- ☒ B. Only 10% of the ^{40}Ar detected in the sample should be used in the calculation. [47%]
- ☐ C. It must be assumed that no ^{40}Ar was present in the rock when it originally solidified. [15%]
- ☐ D. It must be assumed that no ^{40}Ar has escaped from the rock since it

solidified. [11%]

Correct

48% answered correctly

Explanation:



Scientific methods for making measurements and calculations **always have limitations**. When making measurements or calculations, **certain assumptions must be made**. Often the accuracy of these assumptions can be validated or refuted by additional experiments.

The passage describes a method for dating the age of an igneous rock based on the ratio of potassium-40 (^{40}K) to argon-40 (^{40}Ar) found within that rock. This method relies on the observation that gases such as **argon can escape from molten rock** but are trapped within solid rocks. Therefore, any ^{40}K that decays to ^{40}Ar while the rock is molten is expected to escape but **after the rock solidifies**, any new ^{40}Ar produced is expected to be trapped.

However, it is possible that not all the ^{40}Ar that formed while the rock was molten actually escaped, making the rock appear older than it really is. Similarly, it is possible that leaks in the rock allowed some ^{40}Ar to escape after solidification, making the rock appear younger than it really is. Consequently, researchers calculating the age of the rock must **assume that all ^{40}Ar escaped prior to solidification, and no ^{40}Ar escaped afterward**. This assumption could be validated if other methods of dating (each with their own limitations) yield the same age (**Choices C and D**).

The passage also notes that 90% of ^{40}K decay results in the production of calcium-40 (^{40}Ca), and only 10% yields ^{40}Ar . Therefore, when calculating the age of a rock by the $^{40}\text{K} : ^{40}\text{Ar}$ ratio, the ^{40}Ar detected must be assumed to represent only 10% of the original ^{40}K , and only **10% of the remaining ^{40}K should be used** to determine the

$^{40}\text{K} : ^{40}\text{Ar}$ ratio **(Choice A)**.

The question asks for the consideration that does NOT need to be made when dating the rock. If only 10% of the resulting ^{40}Ar were counted, it would yield a $^{40}\text{K} : ^{40}\text{Ar}$ ratio that is too small, **so this consideration should not be made**.

Educational objective:

The limitations of an experimental method must be considered and accounted for when making scientific measurements and calculations.

Time spent:0 sec

Subject:

General Chemistry

QID:404025

System:

Atomic Theory and Chemical Composition

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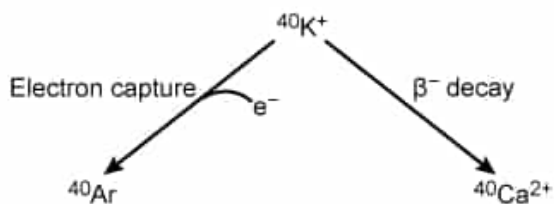


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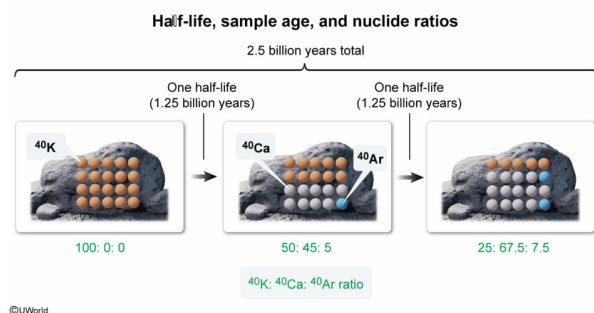
A particular igneous rock sample is dated to an age of 2.5 billion years. A researcher claims that this particular rock contained ^{40}K but no ^{40}Ca when it originally formed. What is the most likely $^{40}\text{K} : ^{40}\text{Ca} : ^{40}\text{Ar}$ ratio observed by the researcher?

- ✔ ☒ A. 25 : 67.5 : 7.5 [60%]
☐ B. 50 : 45 : 5 [14%]
☐ C. 75 : 22.5 : 2.5 [12%]
☐ D. 92.5 : 0 : 7.5 [12%]

Correct

59% answered correctly

Explanation:



The **half-life** of a radioactive **isotope** is the amount of time required for 50% of the isotopes in a sample to decay to their **daughter nuclides**. As such, the age of a sample containing radioactive isotopes can be measured in half-lives. After one half-life, 50% of the original amount of parent nuclides remains; after two half-lives, 25% of the original amount remains; after three half-lives, 12.5% remains, and so on.

The passage states that the half-life of ^{40}K is 1.25 billion years. Accordingly, a rock that is dated at 2.5 billion years old has undergone two half-lives, and 25% of the original amount of ^{40}K remains (ie, 75% has decayed). The passage further states that of the daughter nuclides formed, 90% are ^{40}Ca and 10% are ^{40}Ar . Consequently, of the 75% of ^{40}K nuclides that have decayed, 10% have become ^{40}Ar for a total of 7.5% of the nuclides ($10\% \times 75\% = 7.5\%$). The remaining 67.5% have been converted to ^{40}Ca .

Although the passage mentions that many rocks contain ^{40}Ca from sources other than ^{40}K decay, the researcher in this question claims that this particular rock contained no ^{40}Ca when it originally solidified. Therefore, if the researcher is correct, the only source of ^{40}Ca in this sample is ^{40}K decay, and **the ratio of $^{40}\text{K} : ^{40}\text{Ca} : ^{40}\text{Ar}$ must be 25 : 67.5 : 7.5**.

(Choice B) A ratio of 50 : 45 : 5 would be observed after one half-life (1.25 billion years), not two half-lives.

(Choice C) A ratio of 75 : 22.5 : 2.5 might be obtained by incorrectly using 75% as the amount of ^{40}K that *remained* rather than the amount of ^{40}K that decayed.

(Choice D) A ratio of 92.5 : 0 : 7.5 assumes that no ^{40}Ca was produced by radioactive decay. The question only states that no ^{40}Ca was *originally* present, not that ^{40}Ca was never subsequently produced.

Educational objective:

The half-life of a radioactive isotope is the time required for half the isotopes in a sample to decay to their daughter nuclides. If the half-life and age of a sample are

known, the ratio of parent to daughter nuclides can be calculated.

Time spent:0 sec

Subject:

General Chemistry

QID:404026

System:

Atomic Theory and Chemical Composition

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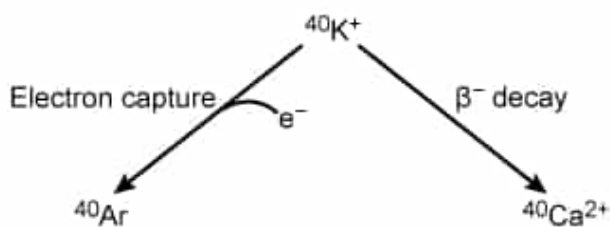


Figure 1 Schematic of the two modes of ^{40}K decay with potassium in the form of a K^+ ion

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Which of the following describes how the classification of a ^{40}K nucleus changes when it undergoes electron capture?

- ☐ A. It is converted from an alkali metal to an alkaline earth metal. [10%]
- ☐ B. It is converted from an alkali metal to a transition metal. [1%]
- ☐ C. It is converted from an alkali metal to a halogen. [4%]

✔ ☒ D. It is converted from an alkali metal to a noble gas. [83%]

Correct

82% answered correctly

Explanation:

Radioactive decay and element classification

1 H 1.0																	2 He 4.0				
3 Li 6.9	4 Be 9.0															5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2
11 Na 23.0	12 Mg 24.3															13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.5	18 Ar 39.9
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.4	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8				
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 95.9	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3				
55 Cs 132.9	56 Ba 137.3	57 La* 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0					
87 Fr (223)	88 Ra (226)	89 Ac* (227)	90 Th (232)	91 Pa (231)	92 U (238)	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)					

Alkali metal

Noble gas

Electron capture

Decay: $^{40}_{19}\text{K} \rightarrow ^{40}_{18}\text{Ar}$

The periodic table groups elements into rows and columns based on their **electron configurations**. Elements that have the same **valence shell** in their neutral form are all in the same row (eg, H and He have 1s valence shells, so they are both in the first row). Elements that have the **same number of electrons** in their valence shell are all in the same column (eg, Li and Na both have a single electron in a valence s orbital, so they are both in the first column). Because elements in a given column have similar chemical properties, they are given a **group name**.

Potassium (K) is found in the first column of the periodic table and, as such, is classified as an alkali metal. According to the passage, when ^{40}K undergoes **electron capture**, it is converted to ^{40}Ar . Argon (Ar) is in the last column of the periodic table because it has a full valence shell in its neutral form, so it is classified as a noble gas. Therefore, when ^{40}K undergoes electron capture, it is converted from an alkali metal to a noble gas.

(Choice A) Alkaline earth metals are found in the second column of the periodic table. ^{40}K is converted to an alkaline earth metal (^{40}Ca) when it undergoes **β^- decay**, not electron capture.

(Choice B) The transition metals are found in the **d-block** of the periodic table. Neither mode of nuclear decay discussed in the passage can convert

^{40}K into a transition metal.

(Choice C) Halogens are found in the second-to-last column of the periodic table. ^{40}K would be converted to a halogen (^{36}Cl) if it underwent α decay, not electron capture. The passage does not indicate that ^{40}K ever undergoes α decay.

Educational objective:

Elements are classified by their electron configurations. Elements in the same column of the periodic table have similarly filled valence shells in their neutral state, similar chemical properties, and a group name. Nuclear decay changes the identity of an atom, and therefore changes its classification.

Time spent:0 sec

QID:404027

Subject:
General Chemistry

System:
Atomic Theory and Chemical Composition

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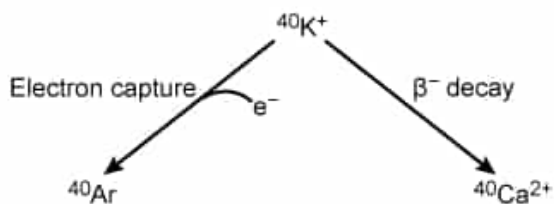


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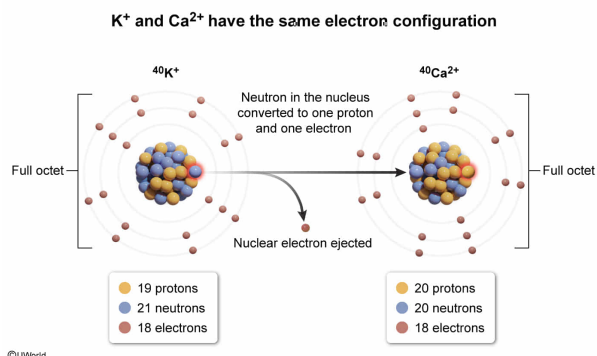
Suppose a $^{40}\text{K}^+$ ion undergoes β^- decay. Which of the following changes to the electron configuration of the resulting isotopes is most likely to occur after the decay?

- ☐ A. It will quickly lose an electron from the 4s shell to the environment to restore a full octet in the valence shell. [30%]
- ☐ B. An electron from the environment will quickly enter the 4s shell to restore a full octet in the valence shell. [22%]
- ☐ C. It will quickly lose two electrons from the 4s shell to the environment to restore a full octet in the valence shell. [18%]
- ☒ D. The electron configuration will not change because the valence shell contains a full octet both before and immediately after the decay. [27%]

Correct

27% answered correctly

Explanation:



The **valence shell** of an atom is generally stabilized when all its suborbitals are fully occupied (ie, the atom has a full octet). Neutral metal atoms commonly **lose electrons** to empty their valence shell, causing the next lowest shell (which contains a full octet) to become the new valence shell. Because this process involves the loss of negative charge, **metal ions are positively charged**.

The question asks about a $^{40}K^+$ ion, which is a ^{40}K atom that has lost a single electron. Because **neutral potassium has a single electron** in its **valence shell**, the **loss of one electron gives it a full octet**. When $^{40}K^+$ decays by **β^- decay**, it loses another electron. However, this electron **does not come from one of the orbitals**. Instead, it is produced when **a neutron in the nucleus is transformed into one proton and one electron**. The newly formed electron is ejected with high energy, and the newly formed proton remains in the nucleus.

The nucleus now has one additional proton and one less neutron, converting it into a calcium-40 (^{40}Ca) nucleus, but the electrons *surrounding* the nucleus (ie, the orbital electrons) have not changed. Consequently, the $^{40}K^+$ ion is transformed into a $^{40}Ca^{2+}$ ion. This form of calcium is electronically stable because, like $^{40}K^+$, $^{40}Ca^{2+}$ has a full octet. Consequently, the **$^{40}Ca^{2+}$ ion will not gain or lose any orbital electrons because it already has a full octet**.

(Choices A and C) The $^{40}Ca^{2+}$ ion produced by the β^- decay does not have any electrons in its 4s shell, so it cannot lose any electrons from this shell. *Neutral* calcium (Ca) commonly **loses two electrons** from the 4s shell to become Ca^{2+} , but Figure 1 shows that the ion formed by β^- decay in this case is *already* Ca^{2+} and does not need to lose any orbital electrons to gain a full octet.

(Choice B) If the $^{40}Ca^{2+}$ ion produced by β^- decay gained an electron in its 4s orbital, it would become $^{40}Ca^+$, which does *not* have a full octet in its valence shell.

Educational objective:

Neutral atoms tend to lose or gain as many electrons as needed to achieve a full octet.

Time spent:0 sec

Subject:

General Chemistry

QID:404028

System:

Atomic Theory and Chemical Composition

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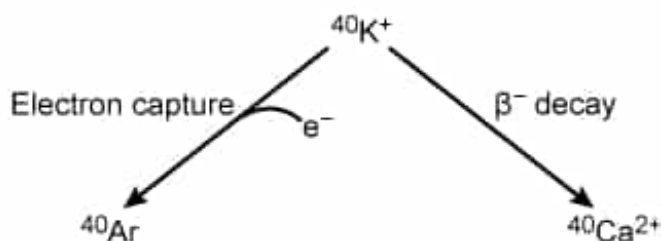


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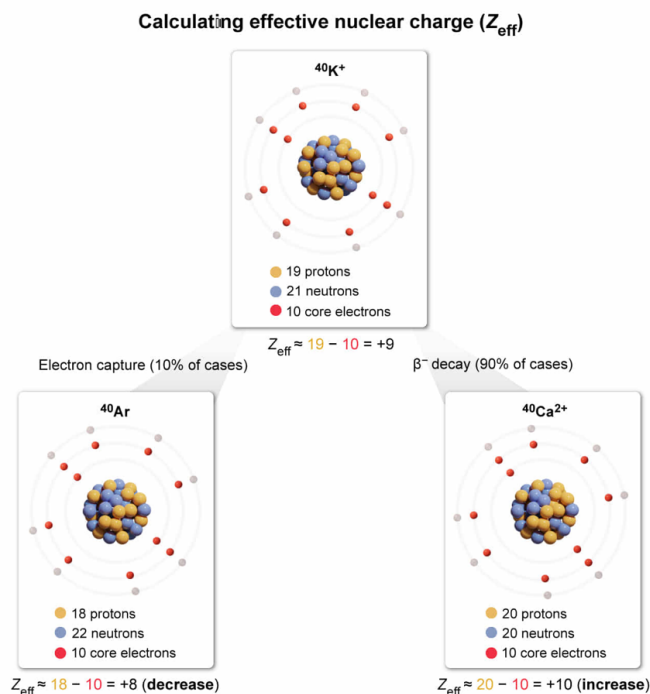
Assuming all atoms return to the ground state immediately after nuclear decay, which of the following best describes the change in effective nuclear charge when a $^{40}\text{K}^+$ ion undergoes nuclear decay? The effective nuclear charge:

- ☐ A. decreases in 90% of cases and increases in 10% of cases. [19%]
- ✓ ☒ B. increases in 90% of cases and decreases in 10% of cases. [48%]
- ☐ C. increases regardless of the mode of decay. [24%]
- ☐ D. decreases regardless of the mode of decay. [7%]

Correct

49% answered correctly

Explanation:



Effective nuclear charge (Z_{eff}) is a measure of the amount of charge from the nucleus of an atom or ion that exerts electrostatic attraction on its valence electrons. **Core electrons** partially shield the valence electrons from the nuclear charge, so the nuclear **charge experienced by the valence electrons is typically less than the total charge of the nucleus**. Z_{eff} for a specific atom or ion is calculated as:

$$Z_{\text{eff}} = Z - S$$

where Z is the total charge of the nucleus (ie, the total number of protons) and S is the shielding constant. S can be approximated as the number of core electrons in the atom or ion. Potassium contains 19 protons and K^+ contains 18 electrons. Because K^+ has an empty 4th shell, its 3rd shell is its valence shell, which means K^+ has 8 valence electrons and 10 core electrons. Based on this count, the effective nuclear charge of any isotope of K^+ is approximately:

$$Z_{\text{eff}} \approx 19 - 10 = +9$$

According to the passage, 90% of decay events are β^- decay, and Figure 1 shows that this results in a Ca^{2+} ion. Ca^{2+} has 20 protons and 18 electrons, 10 of which are core electrons. Consequently, for Ca^{2+} , Z_{eff} is approximately:

$$Z_{\text{eff}} \approx 20 - 10 = +10$$

Therefore, **in the 90% of decay events that produce Ca^{2+} , Z_{eff} increases from +9 to +10.**

The remaining 10% of decay events occur through electron capture, which produces neutral argon (Ar) according to Figure 1. Ar contains 18 protons and 18 electrons, 10 of which are core electrons. Consequently, for Ar, Z_{eff} is approximately:

$$Z_{\text{eff}} \approx 18 - 10 = +8$$

Therefore, **in 10% of decay events, Z_{eff} decreases from +9 to +8.**

(Choices A, C, and D) K^+ has a smaller Z_{eff} than Ca^{2+} but a greater Z_{eff} than Ar. Therefore, Z_{eff} must *increase* in the 90% of decay events that produce Ca^{2+} (ie, β^- decay) and decrease in the 10% of cases that produce Ar (ie, electron capture).

Educational objective:

Effective nuclear charge (Z_{eff}) can be approximated by the equation:

$$Z_{\text{eff}} = Z - S$$

where Z is the number of protons in the nucleus and S is the number of core electrons.

Time spent:0 sec

Subject:
General Chemistry

QID:404029

System:
Atomic Theory and Chemical Composition